

OPPONG KWABENA ADUTWUM

COMPUTER SCIENCE UNDERGRADUATE

Applied ML · Robotics · Embedded Intelligence · Full-Stack Development

CONTACT

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github.com/Oppong-py
oppong-py.github.io

TECHNICAL SKILLS

Programming Languages

- Python
- C++

Data & Analytics

- NumPy / pandas
- SQL
- Jupyter Notebooks
- Matplotlib / Seaborn
- Preprocessing & EDA

AI & ML Tools

- Scikit-learn
- Groq AI API
- Claude Code / Claude Cowork

Web & Backend

- React Native / Expo

Dev Tools & Design

- Git / GitHub
- VS Code
- Figma
- Linux (Kali)

Core Knowledge

- Machine Learning Fundamentals
- Transfer Learning
- Cross-validation & Leakage
- PID Control / Robotics
- Data Structures & Algorithms
- OSI & TCP/IP Networking

LANGUAGES

- English — Native
- French — Intermediate
- Twi — Native

CERTIFICATIONS

- Certificate of Participation — Ghana Data Science Regional Hackathon (Sep 2026)
- Hashgraph Developer Course — Hedera (Jan 2026)
- Hedera Certified Foundation (Apr 2026)
- IPPF Competitor — Brewer Foundation & NYU (2024-25)
- WASSCE General Science — Prempeh College (Sep 2025)

GITHUB HIGHLIGHTS

- [gds-hackathon-2026-building-age](#)
- [ml-specialization-coursera](#)
- [oppong-py.github.io](#)

INTERESTS

- Mathematics
- Machine Learning & AI
- Robotics & Embedded Systems
- UI/UX & Visual Design
- Movies & Video Games
- Sports (Football)

MEMBERSHIPS

- Prempeh College Robotics Club (2023–2025)
- AmaliTech Coding Club (2025–Present)
- Microsoft User Group — KNUST (2025–Present)
- Claude Builder Club — KNUST (2025–Present)

PROFESSIONAL SUMMARY

Second-year Computer Science student at KNUST. Started with robotics, went through network security, and has landed on applied machine learning. National Robotics Champion (Robofest Ghana 2025) and Regional Cybersecurity Champion (2024).

EXPERIENCE

Machine Learning Trainee — ZapTek Research and Innovation Hub

Remote | May 2026

- Completed a 2-week introductory ML programme covering data collection, preprocessing, model training and evaluation; currently in the second cohort.
- Cleaned datasets using Pandas and NumPy, handling missing values and feature scaling.
- Trained classification models with Scikit-learn and compared performance using accuracy, precision, recall, and F1-score.

Online Safety Analyst & Volunteer — Solutions for Life Initiative Ghana (SFLIG)

Ghana | Oct 2022 – Present

- Joined as a volunteer (Oct 2022); promoted to Online Safety Analyst (Aug 2023), managing social media and the website with a focus on internet security.
- Runs speaking engagements with girls from orphanages and rural communities on online safety topics.

Co-Founder & Software Developer — Arethos Tech Initiatives

Kumasi, Ghana | 2026 – Present

- Co-founding and leading development of Arethos, a mental wellness app for KNUST students: technical roadmap, AI-assisted features, and a small dev team.
- Won 2nd Place at ACES CodeFest 2026 UI/UX Challenge. Redesigned the ACES KNUST website for mobile across all 9 pages as a team of 5, building a clickable Figma prototype.

EDUCATION

Kwame Nkrumah University of Science & Technology (KNUST)

B.Sc. Computer Science | Kumasi, Ghana | Expected 2029

- Coursework: Networking & Internet (OSI/TCP/IP, IPv4, subnetting), Calculus I, Probability & Statistics, C++.

Prempeh College (Senior High School)

WASSCE — General Science | Kumasi, Ghana | Graduated September 2025

PROJECTS

Arethos — Mental Wellness App (In Development)

React Native · Expo SDK 54 · SQLite · Groq AI

- Cross-platform mental wellness app with AI-assisted journaling, mood tracking, and adaptive reflections. Built with local-first storage and Groq AI integration for personalized responses.

Autonomous Line-Following Robot

LEGO MINDSTORM EV3 · PID Control Algorithm

- Designed and hand-tuned a PID control algorithm for autonomous line-following; National Champion at Robofest Ghana 2025.
- Tuned the sensor-motor feedback loop by iteration — run, watch the oscillation, adjust, run again.

Solar-Powered Automated Water Dispenser

Arduino · Ultrasonic Sensors · C++ · Solar

- Built a solar-powered water dispensing system using dual ultrasonic sensors to detect container placement and fill level, automatically controlling a solenoid valve.
- Implemented debounce logic, buzzer feedback, LED status indicators, and off-grid solar power for deployment without reliable electricity.

Class Attendance Monitoring System

React Native · Django · PostgreSQL · Bluetooth Low Energy

- Built a biometric attendance platform and tested it with a class of 50 students.
- Built a Django REST API handling concurrent Bluetooth fingerprint scans, including the race conditions that come with them.

HACKATHONS & COMPETITIONS

Ghana Data Science Regional Hackathon 2026 — Participant — Machine Learning (Sept 2026)

- Building-age classification from 40 years of Landsat imagery, Madrid to Amsterdam; team of four. Found the supplied evaluation inflated by 0.047 macro-F1 through spatial leakage. github.com/Oppong-py/gds-hackathon-2026-building-age

ACES CodeFest 2026 — 2nd Place — UI/UX Design (July 2026)

- Redesigned the ACES KNUST website for mobile across all 9 pages as a team of 5, as a clickable Figma prototype built with real-time collaboration.

Robofest Ghana — National Champion — Robotics (2025)

- Built autonomous robots under timed conditions to complete designated tasks; designed and assembled within strict time limits.

National Cybersecurity Challenge — Regional Champion (2024)

- A written challenge on the basics of cyber defence: threat analysis, network security and good practice. Won the regional round and went through to the national stage.

International Public Policy Forum — Competitor (2024-25)

LEADERSHIP & VOLUNTEERING

Club Leader — Prempeh College Cybersecurity Club (2023–2025)

President — House Christian Fellowship, Prempeh College

Four named references with contact details: oppong-py.github.io/testimonials.html